

Agentic AI: From Prompting to Systems That Act

Reuben Brasher

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Lecture 7

The Problem with Agentic AI

Most discussions of agentic AI ignore necessary system constraints.

- ▶ Agents are framed as autonomous actors.
- ▶ Narratives emphasize independence.
- ▶ Real systems require boundaries.
- ▶ Unconstrained autonomy amplifies risk.
- ▶ LLM unreliability persists.
- ▶ Autonomy without control is dangerous.

Bridge from Failure to Agency

Agentic systems inherit all prior failure modes.

- ▶ Hallucination persists in agents.
- ▶ Structural inconsistency compounds.
- ▶ Context drift worsens.
- ▶ Silent errors propagate.
- ▶ Failures become harder to detect.
- ▶ Complexity increases instability.

What Is Agentic AI

Agentic AI performs structured multi-step actions.

- ▶ Maintains state across interactions.
- ▶ Executes sequences of tasks.
- ▶ Interacts with external systems.
- ▶ Coordinates tools and processes.
- ▶ Operates over workflows.
- ▶ Transforms inputs across stages.

From Sequences to Systems

Agentic AI extends simple transformations into systems.

- ▶ Prompts become pipelines.
- ▶ Conversations become workflows.
- ▶ Outputs feed forward.
- ▶ State accumulates.
- ▶ Systems become iterative.
- ▶ Complexity increases.

The Illusion of Autonomy

Claims of autonomy ignore known limitations.

- ▶ Models do not understand tasks.
- ▶ They cannot validate correctness.
- ▶ They hallucinate under uncertainty.
- ▶ They drift over time.
- ▶ They require constraints.
- ▶ Autonomy amplifies weakness.

Reality Check from Practice

Empirical results contradict autonomy claims.

- ▶ Unstructured systems are brittle.
- ▶ Logical errors persist.
- ▶ Systems degrade over time.
- ▶ Failures are silent.
- ▶ Correction requires control.
- ▶ Reliability is conditional.

Amplified Weakness

Agents amplify underlying model limitations.

- ▶ More steps increase failure points.
- ▶ Errors propagate across stages.
- ▶ Feedback loops reinforce mistakes.
- ▶ Context growth destabilizes.
- ▶ Validation is absent.
- ▶ Complexity multiplies risk.

Agents as Workflows

Agentic systems are structured workflows.

- ▶ Tasks are decomposed.
- ▶ Outputs are structured.
- ▶ Steps are sequenced.
- ▶ Transformations are explicit.
- ▶ State transitions are controlled.
- ▶ Workflows replace prompts.

From Prompting to Pipelines

The shift is toward engineered processes.

- ▶ Natural language defines domain.
- ▶ Domain becomes schema.
- ▶ Schema becomes artifacts.
- ▶ Artifacts are validated.
- ▶ Each step constrains output.
- ▶ Structure ensures reliability.

Role-Based Architecture

Roles define system behavior across stages.

- ▶ Architect defines structure.
- ▶ Generator produces output.
- ▶ Validator checks results.
- ▶ Reviewer evaluates quality.
- ▶ Orchestrator coordinates.
- ▶ Roles enforce separation.

Multi-Agent Patterns

Multiple agents simulate structured collaboration.

- ▶ Writer produces content.
- ▶ Editor refines output.
- ▶ Reviewer evaluates correctness.
- ▶ Architect defines constraints.
- ▶ Implementer executes tasks.
- ▶ Roles operate independently.

Pair Programming Analogy

Agent systems resemble collaborative workflows.

- ▶ Human acts as supervisor.
- ▶ Model acts as executor.
- ▶ Feedback improves output.
- ▶ Errors are caught early.
- ▶ Iteration refines results.
- ▶ Supervision is required.

Reliability Through Separation

Separation enables system stability.

- ▶ Roles have limited scope.
- ▶ Outputs are constrained.
- ▶ Errors are isolated.
- ▶ Validation is independent.
- ▶ Feedback loops are controlled.
- ▶ Structure manages complexity.

Failure Modes in Agents

Agent systems introduce compound failures.

- ▶ Cascading hallucinations occur.
- ▶ Incorrect assumptions reinforce.
- ▶ Feedback loops go unchecked.
- ▶ Context grows excessively.
- ▶ Tests validate errors.
- ▶ Traceability is lost.

Feedback Loop Risk

Agents can amplify their own errors.

- ▶ Outputs become inputs.
- ▶ Errors embed in state.
- ▶ Corrections are not applied.
- ▶ Confidence increases falsely.
- ▶ Iteration misleads.
- ▶ Degradation appears as progress.

Context Explosion

Uncontrolled context destabilizes systems.

- ▶ Irrelevant information accumulates.
- ▶ Noise increases.
- ▶ Signal clarity decreases.
- ▶ Objectives drift.
- ▶ Hallucination risk rises.
- ▶ Reliability declines.

The Missing Layer: Control

Agentic systems require explicit control.

- ▶ Constraints must be enforced.
- ▶ Validation must be external.
- ▶ State must be managed.
- ▶ Boundaries must be defined.
- ▶ Outputs must be verified.
- ▶ Autonomy must be limited.

Control as Design

Control is an architectural requirement.

- ▶ Interfaces must be defined.
- ▶ Data flow must be restricted.
- ▶ Determinism should be enforced.
- ▶ State transitions tracked.
- ▶ Validation checkpoints added.
- ▶ Auditability maintained.

Design Principles

Effective systems follow strict design rules.

- ▶ Constrain outputs at each stage.
- ▶ Isolate contexts.
- ▶ Require validation.
- ▶ Limit component scope.
- ▶ Avoid large tasks.
- ▶ Include human checkpoints.

Human-in-the-Loop

Human oversight ensures system reliability.

- ▶ Validate intermediate outputs.
- ▶ Correct errors early.
- ▶ Guide system direction.
- ▶ Enforce constraints.
- ▶ Provide domain knowledge.
- ▶ Prevent silent failures.

Structural Parallel

System architecture reflects these principles.

- ▶ Bounded contexts enforce separation.
- ▶ Event sourcing provides traceability.
- ▶ Deterministic interfaces improve reliability.
- ▶ Randomness is externalized.
- ▶ Orchestration is controlled.
- ▶ Components remain independent.

Key Insight

Agentic AI is about controlled execution.

- ▶ Behavior must be constrained.
- ▶ Workflows must be defined.
- ▶ Roles must be separated.
- ▶ Outputs must be validated.
- ▶ State must be managed.
- ▶ Control must be continuous.

Transition to Governance

Agent systems operate on data and actions.

- ▶ They process inputs and outputs.
- ▶ They interact with systems.
- ▶ They affect data flows.
- ▶ They introduce risk.
- ▶ They require accountability.
- ▶ They must be governed.

Bridge to Lecture 8

Agentic systems must align with enterprise constraints.

- ▶ They must follow data governance.
- ▶ They must integrate with systems.
- ▶ They must support auditability.
- ▶ They must manage data lifecycle.
- ▶ They must respect policy.
- ▶ They become infrastructure.

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